

# Mathematical and Computational Methods

## Exercise Sheet 9: Integration I: the definite integral and substitution

This sheet accompanies *Unit 9 (Integration I: the definite integral and substitution)*. It exercises Riemann sums and signed area, both parts of the Fundamental Theorem of Calculus, and integration by substitution; no technique from Unit 10 (parts, partial fractions, trigonometric substitution, improper integrals) is needed.

**How to use this sheet.** Problems are graded by difficulty—★ drill (routine fluency), ★★ core (standard, tutorial level) and ★★★ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ►) and **further practice** for independent home study.

**Solutions.** A full worked solution sits after each problem: hidden in the student build, shown in the solutions build. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showssolutions{}\input{MCM_Exercises_Unit9}"`) and recompile.

**Toolkit.** Riemann sums and signed area; the power rule and the table of standard integrals; both parts of the Fundamental Theorem of Calculus (FTC); and integration by substitution, including changing the limits and the  $f'/f \rightarrow \ln$  pattern.

### 1 Warm-up drills

Short, single-idea problems — at least one for every skill in the unit.

#### Riemann sums and signed area

**Exercise 1.1** (★ drill — *a right-endpoint Riemann sum*). Estimate the area under  $f(x) = x^2$  on  $[0, 2]$  using  $n = 4$  rectangles of equal width and the height at the *right* end of each strip. Is your estimate an over- or under-estimate, and why?

**Exercise 1.2** (★ drill — *the sign of an integral*). Without computing any antiderivative, decide the sign (positive, negative or zero) of each definite integral, and say why.

$$(a) \int_{-2}^2 x^3 \, dx, \quad (b) \int_0^1 e^{-x} \, dx, \quad (c) \int_{\pi}^{2\pi} \sin x \, dx.$$

**Exercise 1.3** (★ drill — *sketching the rectangles*). A picture first.

- Sketch  $y = x^2$  on  $[0, 2]$  and, on the same axes, draw the four right-endpoint Riemann rectangles of Exercise 1 (width 0.5). Mark whether the rectangles lie above or below the curve.
- On separate axes sketch  $g(x) = x - 1$  on  $[0, 2]$ . Shade the region between the graph and the  $x$ -axis, marking the part that counts as *negative* signed area. Then compute the net signed area  $\int_0^2 (x - 1) \, dx$  and the total (unsigned) area.

## Properties of the definite integral

**Exercise 1.4** (*★ drill — properties of the definite integral*). Suppose  $\int_0^2 f = 5$  and  $\int_2^5 f = -2$ . Using only the properties of the definite integral (do *not* find  $f$ ), evaluate

$$(a) \int_0^5 f, \quad (b) \int_5^2 f, \quad (c) \int_2^2 f, \quad (d) \int_0^2 3f, \quad (e) \int_5^0 f.$$

## Indefinite integration and the power rule

**Exercise 1.5** (*★ drill — a polynomial antiderivative*). Find  $\int (x^3 - 4x + 2) \, dx$ .

**Exercise 1.6** (*★ drill — powers of  $x$  first*). Write each integrand as a power of  $x$  first:  $\int \left( \sqrt{x} + \frac{1}{x^2} \right) \, dx$ .

**Exercise 1.7** (*★ drill — standard integrals*). Use the table of standard integrals:  $\int \left( 2e^x - \frac{3}{x} + 5 \cos x \right) \, dx$ .

## Standard integrals (including arctan/arcsin)

**Exercise 1.8** (*★ drill — standard forms with a linear inside*). Evaluate the indefinite integrals

$$(a) \int e^{3x} \, dx, \quad (b) \int \sec^2(2x) \, dx, \quad (c) \int \frac{dx}{9+x^2}, \quad (d) \int \frac{dx}{\sqrt{4-x^2}}, \quad (e) \int \cos(5x) \, dx.$$

## FTC Part 2: evaluating definite integrals

**Exercise 1.9** (*★ drill — definite integrals by FTC 2*). Evaluate

$$(a) \int_1^3 x^2 \, dx, \quad (b) \int_0^{\pi/2} \cos x \, dx, \quad (c) \int_1^e \frac{dx}{x}, \quad (d) \int_0^1 e^{2x} \, dx.$$

## FTC Part 1: differentiating an integral

**Exercise 1.10** (*★ drill — FTC Part 1*). Differentiate with respect to  $x$  (do *not* try to evaluate the integral):

$$(a) \frac{d}{dx} \int_0^x e^{-t^2} \, dt, \quad (b) \frac{d}{dx} \int_2^x \sin(t^3) \, dt, \quad (c) \frac{d}{dx} \int_1^x \frac{dt}{1+t^4}.$$

## Substitution (reverse chain rule and $f'/f$ )

**Exercise 1.11** (*★ drill — substitution*). Each integrand contains an “inside” function whose derivative is present. Integrate by substitution:

$$(a) \int 2x(x^2+1)^5 \, dx, \quad (b) \int \cos x e^{\sin x} \, dx, \quad (c) \int 3x^2(x^3+7)^4 \, dx.$$

**Exercise 1.12** (*★ drill — the  $f'/f$  pattern*). Spot the  $f'/f$  pattern and write down the logarithm:

$$(a) \int \frac{2x}{x^2+5} \, dx, \quad (b) \int \frac{\cos x}{\sin x} \, dx, \quad (c) \int \frac{e^x}{e^x+1} \, dx.$$

**Exercise 1.13** (*★ drill — changing the limits*). Evaluate  $\int_0^1 x e^{x^2} \, dx$  by changing the limits to the new variable.

## No elementary antiderivative

**Exercise 1.14** (*★ drill — elementary antiderivatives?*). For each function, state whether it has an elementary antiderivative.

$$(a) e^{-x^2}, \quad (b) x e^{-x^2}, \quad (c) \frac{\sin x}{x}, \quad (d) \sin(x^2), \quad (e) x \cos(x^2).$$

## 2 Tutorial problems

The core set for the two-hour tutorial. Standard difficulty, anchored on the worked examples of Unit 9, with one problem flavoured for each cohort. Problems flagged ► are the live core worked together in the two-hour session; the remaining problems are for completion at home.

**Exercise 2.1** (*★★ core — Riemann sums against the exact area*). ► Let  $f(x) = 1 + x^2$  on  $[0, 3]$ , cut into  $n = 3$  strips of width 1.

- Compute the right-endpoint and left-endpoint Riemann sums.
- Using FTC Part 2, find the exact area and check it lies between your two estimates.

**Exercise 2.2** (*★★ core — simplify, then integrate*). ► Integrate, simplifying first where it helps:

$$(a) \int \frac{x^2 - 3x + 2}{x} dx, \quad (b) \int (e^{2x} + \frac{3}{x} - \cos 5x) dx.$$

**Exercise 2.3** (*★★ core — FTC evaluation*). (*FTC evaluation — worked-example anchor*). ► Evaluate  $\int_2^4 x^2 dx$  and  $\int_0^1 (3x^2 - 4x + 5) dx$ , and state in one sentence what the first number represents geometrically.

**Exercise 2.4** (*★★ core — a rate of flow (physics)*). [Physics / applied] Water flows into a tank at the rate  $r(t) = 6t$  litres per minute,  $0 \leq t \leq 5$ . The volume added between times  $a$  and  $b$  is  $\int_a^b r(t) dt$ .

- How much water enters during the first 5 minutes?
- During which single minute  $[t, t + 1]$  does the most water enter, and how much?

**Exercise 2.5** (*★★ core — displacement and distance (physics)*). [Physics / applied] A particle moves on a line with velocity  $v(t) = 3t^2 - 12t + 9$  (m s<sup>-1</sup>) for  $0 \leq t \leq 3$ .

- Find the net displacement  $\int_0^3 v dt$ .
- Find the total distance travelled,  $\int_0^3 |v| dt$ .

**Exercise 2.6** (*★★ core — substitution with a limit change*). (*Substitution with a limit change — worked-example anchor*). ► Evaluate  $\int_0^2 x\sqrt{x^2 + 1} dx$  by converting the limits to the new variable.

**Exercise 2.7** (*★★ core — a stray x to rewrite*). Evaluate  $\int_0^3 x\sqrt{x + 1} dx$ . (Here the substitution leaves a stray  $x$  behind — rewrite it in terms of  $u$ .)

**Exercise 2.8** (*★★ core — two routes to a logarithm*). ► Evaluate two integrals, contrasting the methods:

$$(a) \text{ by the } f'/f \text{ pattern, } \int \tan x dx, \quad (b) \text{ by substitution, } \int_1^2 \frac{\ln x}{x} dx.$$

**Exercise 2.9** (★★ core — completing the square). ▶ By completing the square, evaluate  $\int \frac{dx}{x^2 + 2x + 2}$ .

**Exercise 2.10** (★★ core — a probability density (data science)). [Data science / probability] A continuous random variable  $X$  has probability density  $f(x) = kx(1-x)$  on  $[0, 1]$  and  $f(x) = 0$  elsewhere. A density must satisfy  $\int_0^1 f = 1$ , and  $\mathbb{P}(X \leq c) = \int_0^c f$  (probability is area).

- Find  $k$ .
- Find  $\mathbb{P}(X \leq \frac{1}{2})$  and comment on the answer.

**Exercise 2.11** (★★ core — FTC 1 with the chain rule). [Mathematics / pure] ▶ Let  $g(x) = \int_0^{x^2} \cos t \, dt$ . Find  $g'(x)$  in two different ways and check they agree.

**Exercise 2.12** (★★ core — odd and even on a symmetric interval (maths)). [Mathematics / pure] ▶ Let  $f$  be continuous on  $[-a, a]$ . Using the substitution  $u = -x$  on  $\int_{-a}^0 f$ , prove that  $\int_{-a}^a f(x) \, dx = 0$  whenever  $f$  is odd, and that it equals  $2 \int_0^a f(x) \, dx$  whenever  $f$  is even.

**Exercise 2.13** (★★ core — spot the error). (Spot the error / conceptual.)

- A student writes  $\int_{-1}^1 \frac{dx}{x^2} = \left[-\frac{1}{x}\right]_{-1}^1 = (-1) - (1) = -2$ . The integrand is positive everywhere it is defined, so a negative “area” is impossible. What went wrong?
- True or false: “ $\int_a^b f(x) \, dx$  is a function of  $x$ .” Justify.

### 3 Further practice (home)

A larger bank for independent study, running from routine to harder. Challenge and extension problems are marked ★ ★ ★.

#### Indefinite integrals

**Exercise 3.1** (★ drill — indefinite drill). Find

- $\int (4x^3 - 6x^2 + 1) \, dx$ , (b)  $\int \left(\frac{2}{x^3} + 5\sqrt{x}\right) \, dx$ , (c)  $\int \left(e^{-x} + \frac{1}{1+x^2}\right) \, dx$ , (d)  $\int (3 \sin x - 4 \sec^2 x) \, dx$ .

#### Definite integrals (FTC Part 2)

**Exercise 3.2** (★ drill — definite drill). Evaluate

- $\int_1^2 \frac{dx}{x^2}$ , (b)  $\int_0^{\ln 2} e^x \, dx$ , (c)  $\int_0^{\pi/4} \sec^2 x \, dx$ , (d)  $\int_{-1}^1 (x^3 + x) \, dx$ , (e)  $\int_0^4 \sqrt{x} \, dx$ .

#### Substitution — indefinite

**Exercise 3.3** (★ drill — substitution drill). Integrate

- $\int x^2(x^3 + 1)^7 \, dx$ , (b)  $\int \sin^4 x \cos x \, dx$ , (c)  $\int \frac{(\ln x)^2}{x} \, dx$ , (d)  $\int x e^{-x^2} \, dx$ .

**Substitution — definite****Exercise 3.4** (★★ core — *definite substitution*). Evaluate

$$(a) \int_1^2 \frac{\ln x}{x} dx, (b) \int_0^{\pi/2} \sin x \cos x dx, (c) \int_0^1 \frac{x}{(1+x^2)^2} dx, (d) \int_0^1 \frac{2x}{x^2+1} dx.$$

**The  $f'/f \rightarrow \ln$  pattern****Exercise 3.5** (★★ core — *the  $f'/f$  pattern*). Integrate

$$(a) \int \frac{3x^2+2}{x^3+2x+1} dx, \quad (b) \int \frac{\sec^2 x}{\tan x} dx.$$

**arctan and arcsin targets****Exercise 3.6** (★★ core — *inverse-trig standard forms*). Evaluate the definite integrals

$$(a) \int_0^{\sqrt{3}} \frac{dx}{1+x^2}, \quad (b) \int_0^1 \frac{dx}{\sqrt{4-x^2}}, \quad (c) \int_0^2 \frac{dx}{4+x^2}.$$

**Substitution leaving an  $x$  behind****Exercise 3.7** (★★ core — *a substitution leaving a stray  $x$* ). Find  $\int \frac{x}{\sqrt{x+4}} dx$ .**Signed area versus total area****Exercise 3.8** (★★ core — *signed area and total area*). For each function find both the net signed area and the total (painted) area over the stated interval.

- (a)  $\sin x$  on  $[0, 2\pi]$ .  
 (b)  $x^2 - 1$  on  $[0, 2]$ .

**Integral equations and applications****Exercise 3.9** (★★ core — *solve for the limit*). Find the positive value of  $a$  for which  $\int_0^a (2x+1) dx = 12$ .**Exercise 3.10** (★★ core — *work done on a spring (physics)*). [Physics / applied] A spring obeys Hooke's law with force  $F(x) = kx$  ( $k = 200 \text{ N m}^{-1}$ ), where  $x$  is the extension. The work done stretching it from  $x = 0$  to  $x = d$  is  $W = \int_0^d F(x) dx$ . Find the work to stretch the spring by 0.1 m.**Exercise 3.11** (★★ core — *a triangular density (data science)*). [Data science / probability]  $X$  has the triangular density  $f(x) = \frac{1}{2}x$  on  $[0, 2]$  (and 0 elsewhere).

- (a) Verify it is a valid density.  
 (b) Find  $P(X \leq 1)$  as an area.

## Challenges and extensions

**Exercise 3.12** (★★★challenge — FTC 1 with the chain rule). Use FTC Part 1 together with the chain rule to differentiate

$$(a) \frac{d}{dx} \int_0^{x^2} e^t dt, \quad (b) \frac{d}{dx} \int_x^{x^2} \ln(1+t^2) dt.$$

**Exercise 3.13** (★★★challenge — completing the square, log plus arctan). A linear numerator over an irreducible quadratic splits into a logarithm plus an arctangent. Evaluate  $\int \frac{x+1}{x^2+4x+5} dx$ . (Hint: complete the square below, and write the numerator as a multiple of the denominator's derivative plus a constant.)

**Exercise 3.14** (★★★challenge — an odd power under a root). Evaluate  $\int \frac{x^3}{\sqrt{x^2+1}} dx$ . (Hint:  $u = x^2 + 1$ , and write  $x^3 = x^2 \cdot x$ .)

**Exercise 3.15** (★★★challenge — spot the error). (Spot the error.) A student evaluates  $\int_0^{\pi/2} \sin^2 x \cos x dx$  thus:

$$\text{“set } u = \sin x; \int_0^{\pi/2} u^2 du = \left[ \frac{1}{3} u^3 \right]_0^{\pi/2} = \frac{1}{3} \left( \frac{\pi}{2} \right)^3 \text{.”}$$

Identify both mistakes and give the correct value.

**Exercise 3.16** (★★★challenge — no elementary antiderivative (data science)). [Data science / numerical] The integral  $\int_0^1 e^{-x^2} dx$  has no elementary antiderivative (it gives a value of the error function).

- Explain why no FTC 2 “plug-in” will evaluate it exactly.
- Bracket its value using left- and right-endpoint Riemann sums with  $n = 2$  ( $\Delta x = \frac{1}{2}$ ). Is the true value inside your bracket?